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JOURNAL AND PODCAST

TAMIM MOHAMED ALI – 21908541

BOSHRA ALALOU – 21908516

Professor : Manuel Pita

10 de fevereiro - 16 de fevereiro

Introduction to Artificial Intelligence

Overview

Title: What is **Artificial Intelligence**?

Instructor: Professor Manuel Pita

Key Topics and Notes

Some Basic Ideas:

- **AI** aims to enable computers to perform tasks that **human minds** can do, such as **reasoning**, **perception**, **motor control**, **categorization**, **planning**, **explanation**, and **prediction**.
- The **Turing Test** evaluates **intelligence**.
- **Intelligence** is **multi-dimensional**.
- **AI** explores the possibility of separating **intelligence** from the **brain**.

Two Main Motivations:

- **Technological:** Advancing **tools** and **systems**.
- **Scientific:** Understanding **intelligence** and **cognition**.

Two Main Purposes:

- **Predictive:** **Forecasting** outcomes.
- **Explanatory:** Understanding **processes**.

Who Uses AI?

- **Computer scientists**, **biologists**, **neuroscientists**, **governments**, **tech companies** (e.g., **Google**, **Facebook**), **mobile devices**, **police**, **social scientists**, **cognitive scientists**, and **psychologists**.

Three Pillars:

- (No specific details provided in the notes.)

The Two Roots of AI:

- **Symbolic AI (Good Old Fashion AI):**
 - Originates from **logic** and **language**.
 - Uses **symbols** to represent **meaning**.
 - Shares roots with **computer science**, pioneered by **Alan Turing**.
 - Dominant in **AI** from the **1970s** to **2000s**.
 - **Rule-based**, **abstract**, and less **flexible**.
- **Connectionist AI:**
 - Inspired by **neural information processing** and **cybernetics**.
 - Investigates **networks of switches** for **math** and **language**.
 - Incorporates **control** and **feedback**, mimicking **biological systems**.
 - Limited by **computing power** until recently; now known as **Deep Learning**.
 - Highly **flexible**, requires extensive **training data**.

Mid-Class Discussion:

- Are humans **symbolic**, **connectionist**, or something else?

The Pioneers:

- **Lady Ada Lovelace**: First programmer.
- **Alan Turing**: **Turing Machine**, foundation of **computer science**.
- **Warren McCulloch** and **Walter Pitts**: **Neural networks**.
- **Claude Shannon**: **Information communication theory**.
- **Norbert Wiener**: **Feedback** and **memory systems**.
- **John McCarthy**: Created **LISP**, the **first AI language**.
- **Allen Newell** and **Herbert Simon**: **General Problem Solver**.

The Turing Machine:

- Defined by a **transition function**: $\delta(S, q): s \in \Sigma \wedge q \in Q$.
- Reads **symbols** and operates based on **states**.
- Represents the **theoretical universe of computation**.
- **Transition Rules Example**:

- $\delta(q_0, a) = (q_0, a, R)$
 - $\delta(q_0, b) = (q_0, b, R)$
 - $\delta(q_0, \square) = (q_0, \square, R)$
 - $\delta(q_1, a) = (q_1, a, L)$
 - $\delta(q_1, b) = (q_1, b, L)$
 - $\delta(q_1, \square) = (q_1, \square, L)$
-

17 de fevereiro - 23 de fevereiro

Turing Machines

Overview

- Title: Máquinas de Turing
- Instructor: Professor Manuel Pita

Key Topics and Notes

- Introduction:
 - Developed by Alan Turing in 1936, Turing Machines are foundational to computer science and AI.
 - They define the limits of computation and algorithms.
 - Represent one origin of AI through symbol manipulation, contrasting with neural networks inspired by the brain.
- Components:
 - Hardware:
 - A tape divided into cells, each holding a symbol.
 - A read/write head that moves left or right.
 - Software:
 - A set of symbols for input/output.
 - Internal states (initial and final).
 - A transition table dictating actions based on state and symbol.
- Example:
 - A Turing Machine to check if a binary string has an even number of zeros.
 - States: q_0 (start), q_1 (intermediate), q_f (final).
 - Transition rules defined for symbols 0, 1, and blank (B).
- Execution Steps:
 1. Begin in q_0 with the head on the first symbol.
 2. Read the symbol and apply the transition rule.

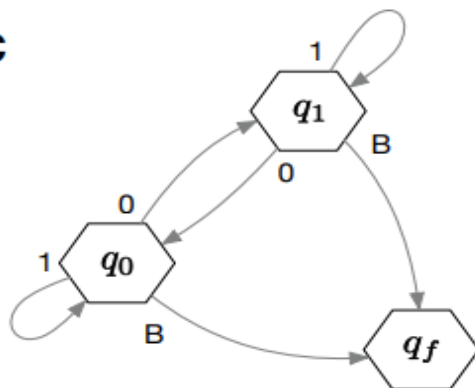
3. Continue until reaching the final state or no transitions remain.
- Philosophical and Practical Context:
 - Address "decidability" — determining which problems can be solved computationally.
 - The Halting Problem is an example of an undecidable problem.
 - Impacts AI by defining computational boundaries.
 - Impact on AI:
 - Provides a theoretical framework for algorithms, from automation to machine learning.
 - Importance:
 - Foundational for understanding computation limits and innovating in AI.
 - Practical Session:
 - Implemented a Turing Machine in Python, defining the program, actions, and state updates.
 - Figures

A

$$S = \{0, 1, B\}$$

$$Q = \{q_0, q_1, q_f\}$$

C



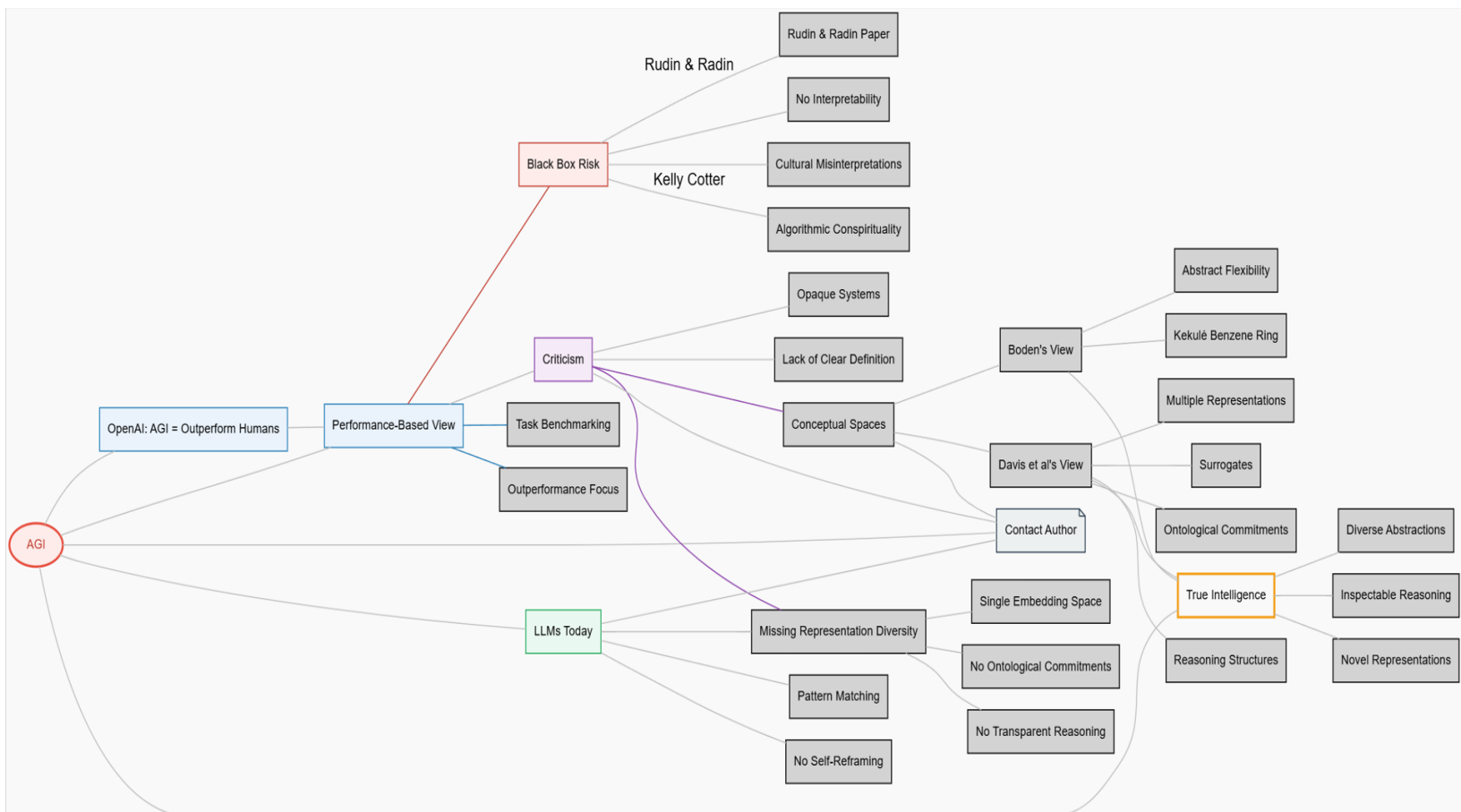
B

Current symbol	Current state	Write	Move	Next state
0	q_0	B	R	q_1
0	q_1	B	R	q_0
0	q_f			
1	q_0	B	R	q_0
1	q_1	B	R	q_1
1	q_f			
B	q_0	0	L	q_f
B	q_1	1	L	q_f
B	q_f		Halt	

[Substack post about Artificial General IntelligenceURL](#)

Professor Manuel Pita argues that the current hype around LLMs achieving human-level intelligence and AGI is wrong because it misdefines what real intelligence is. He criticizes OpenAI's definition of AGI as just outperforming humans in work, saying it ignores key aspects like original thinking and adapting to new situations. This focus on performance leads to a "black box" problem where we don't understand how these AI systems work, making it hard to spot mistakes or biases.

Pita also points out that LLMs lack true understanding. They're good at finding patterns but can't build the diverse and robust concepts that humans can. He uses the example of chemists figuring out benzene's structure to show how humans can change their understanding in ways LLMs currently can't. He emphasizes that true intelligence, as Margaret Boden suggests, involves different kinds of abstract thinking, self-awareness, and the ability to create new ideas by changing how we think. So, just because LLMs can do tasks well doesn't mean they've reached true general intelligence, which requires understanding, explaining, and creating, not just performing.



Podast series: nature of intelligence

Ep. 1: What is Intelligence

What we have is :

- **Human Intelligence and Learning**
Alison Gopnik explains how children develop intelligence by experimenting and understanding cause-and-effect relationships. She emphasizes that this hands-on learning is foundational to human intelligence. John Krakauer adds that much of human intelligence operates unconsciously, particularly in motor skills, where complex actions occur without deliberate thought.
- **Common Sense**
The guests describe common sense as a flexible mix of innate knowledge, learned experiences, and quick inference. This ability allows humans to adapt and navigate diverse situations effectively.
- **The Human Drive to Understand**
A unique aspect of human intelligence is the curiosity to comprehend the world, which has driven advancements like science and technology. Metacognition—thinking about one's own thinking—is highlighted as a critical component of this process.
- **Artificial Intelligence and Large Language Models**
The discussion shifts to whether machines, particularly large language models like ChatGPT, can be considered intelligent. Alison Gopnik uses the "stone soup" metaphor to argue that AI's apparent intelligence stems largely from human input rather than independent reasoning. John Krakauer questions whether machines can achieve true artificial general intelligence, noting they lack feelings and experiential understanding, which he sees as essential for genuine comprehension.
- **Host's Perspective**
Melanie Mitchell recaps the conversation and aligns with the guests' skepticism. She argues that current AI lacks the active interaction with the world and intrinsic motivation necessary for human-like intelligence.

[Ep. 2: The relationship between language and thought](#)

In Ep2 we have >

Part One: Why Do Humans Have Language?

The experts debate the origins of language, challenging Noam Chomsky's idea of an innate "universal grammar." Instead, they argue that language is learned from experience, a view supported by the success of AI language models like ChatGPT. Lupyan posits that language evolved from humans' need to socialize and collaborate, boosting intelligence by enabling knowledge sharing. Piantadosi adds that language efficiently transmits abstract and cultural concepts, though it comes with evolutionary trade-offs like prolonged childhoods.

Part Two: Are Language and Thought Separate in the Brain?

Using fMRI, Fedorenko shows that language and thought activate different brain regions, suggesting they are distinct systems. She points to stroke patients who lose language but retain complex thinking as evidence. Lupyan counters that language might still be essential for developing cognition, particularly in early life. The discussion also covers the "inner voice," which some people lack yet still think effectively, indicating that language isn't always required for thought.

[Episode 3: What kind of intelligence is an LLM?](#)

LLMs work by predicting the next word based on patterns learned from vast text data, which is more statistical than human-like reasoning. They learn differently than humans, lacking real-world experience. While impressive at language tasks, LLMs make basic errors and their "hallucinations" are more like generated hypotheses. They likely lack true beliefs and theory of mind, possibly passing tests through pattern matching. Experts suggest LLMs are on a different path to intelligence than humans, representing a unique but currently limited form of cognition lacking genuine reasoning, self-awareness, and consistent accuracy.

[Episode 4: Babies vs Machines](#)

This episode explores the fascinating ways babies learn and contrasts it with how Large Language Models (LLMs) operate. It highlights that babies, unlike other mammals, spend their early months absorbing a flood of sensory information, primarily seeing hands, objects, and edges. The episode challenges the idea of innate knowledge,

suggesting everything develops through experience. A key contrast is the amount of data: LLMs are trained on trillions of words, while a 5-year-old hears significantly less, yet babies learn more efficiently.

The discussion then delves into *how* babies learn. They selectively focus on interesting and relevant information, learn through multiple senses and physical interaction with their environment (embodied learning), and prioritize communication even before language. Babies excel at "one-shot learning," rapidly generalizing from limited examples, and constantly self-correct their understanding.

The episode argues that babies achieve more with less data due to the quality of their raw, multimodal sensory input, their selective focus, their embodied experiences, social learning, and their adaptive learning mechanisms like rapid generalization and self-correction.

The implications for AI suggest that to become more human-like, AI systems might need to learn from curated, multimodal real-world data, incorporate embodiment, develop selective attention, and build self-evaluation capabilities. The episode concludes by suggesting that while LLMs are useful tools, they don't possess the true intelligence demonstrated by the sophisticated learning processes of babies.

[Episode 5: How do we assess intelligence?](#)

This episode of the Complexity podcast explores how to assess intelligence in non-human entities like animals and AI, noting that human intelligence tests are flawed for this purpose. It traces the history of intelligence testing back to Western philosophy, which often framed intelligence as a human-centric trait tied to reason, impacting how we view other forms of intelligence.

When it comes to **animals**, Erica Cartmill argues against using human language as the benchmark, as it overlooks their unique communication methods (like elephants' infrasonic rumbles). She advocates for species-specific assessments that consider their individual sensory and social worlds.

For **AI, specifically Large Language Models (LLMs)**, Ellie Pavlick discusses how they can pass human tests like the SAT but likely do so by exploiting patterns in the data rather than through genuine reasoning. The "black box" nature of LLMs makes it hard to understand their internal processes, although "mechanistic interpretability" research aims to shed light on this. Pavlick emphasizes that passing human tests doesn't mean LLMs have human-like intelligence.

The episode concludes that our human-centric approach limits our understanding of non-human intelligence. It calls for tailored assessment methods for both animals and AI and stresses the importance of humility and acknowledging the uncertainty in

defining and measuring intelligence across different entities. Better assessment could reveal the cognitive richness of animals and clarify the true capabilities of AI.

Episode 6: AI's changing seasons

Melanie Mitchell, an AI expert, discusses her career and views on the current state of AI in this podcast episode.¹ Inspired by *Gödel, Escher, Bach*, she pursued AI, focusing on analogy-making.² She graduated during an "AI winter," contrasting with today's "AI spring."³

Mitchell notes that while Large Language Models (LLMs) are versatile, they lack true generality and struggle with abstraction and learning from limited examples, as shown by their poor performance on the ARC benchmark.⁴ She is skeptical about existential risks from AI alignment, focusing instead on immediate concerns like deepfakes and misinformation.

Mitchell critiques the AI industry's unsustainable reliance on massive data and energy, advocating for more human-like learning methods, possibly involving embodied AI.⁵ She defines intelligence as the ability to generalize and explains that true understanding involves abstraction, prediction, and explanation, qualities that machines currently lack in a human sense.⁶

Looking to the future, Mitchell envisions AI research shifting towards sustainable and efficient systems, potentially incorporating embodiment. She emphasizes the importance of a multidisciplinary approach, drawing from complexity science to understand the societal and technical implications of AI.⁷

24 de fevereiro - 1 de março

Neural Networks

Overview

- Title: Redes Neurais: Máquinas Feitas de Ligações
- Instructor: Professor Manuel Pita

Key Topics and Notes

- Introduction:
 - Warren McCulloch and Walter Pitts modeled AI after the brain, unlike Turing's symbolic approach.
 - Focused on representation, inference, and learning through neuron-like structures.
 - Biological neurons have dendrites (inputs), axons (outputs), and synapses.
- McCulloch-Pitts Artificial Neuron:
 - Features a threshold (I).
 - Includes excitatory (black circle) and inhibitory (white circle) fibers connecting to the environment or other neurons.
- Dynamics:
 - Operates in discrete time steps.
 - Neurons fire if inputs meet the threshold; inhibitory fibers can block firing.
- Logic Gates and Delays:
 - Can implement AND gates and delay mechanisms using network structures.
- Control and Memory:
 - Control: Inhibitory fibers block signals.
 - Memory: Feedback loops sustain activity.
- Practical Work:
 - Problem 1: Designed a network to control signal flow to two receptors.

- Problem 2: Built a network with feedback for memory (e.g., light on/off).
- Challenge: Analyzed and fixed a binary scaler network's timing issues.
- Solutions:
 - Problem 1: Used inhibitory fibers for selective activation.
 - Problem 2: Employed feedback loops to maintain state.
 - Challenge: Adjusted architecture to resolve timing problems.

2 de março - 8 de março

State Transition Diagrams

Overview

- Recap of Turing Machines and Neural Networks, introducing State Transition Diagrams (STDs).
- Instructor: Professor Manuel Pita

Key Topics and Notes

- State Transition Diagrams (STDs):
 - Represent machine states and transitions.
 - Useful for analyzing and diagnosing behavior.
- Examples:
 - Simple STD with defined transition rules.
 - Single delay neural network diagram.
- How to Draw STDs:
 1. Define machine boundaries.
 2. Identify and represent internal states.
 3. List possible inputs and outputs.
 4. Connect states via inputs/outputs.
 5. Use discrete time steps.
- Complex STDs:
 - Example with multiple inputs and states.
- Key Takeaways:
 1. STDs apply to any machine.
 2. Help diagnose issues.
 3. Reveal reachable/unreachable states.
- Puzzle: Hippies and Punks river crossing problem.

9 de março - 15 de março

Busca não informada

Overview

- Title: Busca não informada
- Instructor: Manuel Pita

Key Topics and Notes

- What is Search in AI?
 - Formalizes problems into states (nodes) and actions (links).
 - Goal: Find a sequence from initial to goal state.
- The Cleaning Robot:
 - Two rooms, connected by a door.
 - Actions: Move between rooms or clean (not both).
 - Represented with states, search graph, and adjacency list.
 - Example action sequences derived from the graph.
- Basic Search Algorithm:
 - Initialize queue with initial state.
 - Dequeue, check for goal; if not, expand and enqueue neighbors.
- Handling Loops:
 - Track visited nodes to avoid cycles.
- Limitations:
 - Tracking nodes alone misses multiple paths; track full paths instead.
- Path-Based Search:
 - Queue stores paths, evolving as search progresses.
- Depth-First vs. Breadth-First Search:
 - DFS: Uses stack (LIFO), explores deep paths, less memory.
 - BFS: Uses queue (FIFO), finds shortest path in unweighted graphs.

- Examples of DFS and BFS step-by-step on a search graph.
- Comparison:
 - BFS: Optimal for minimal actions.
 - DFS: Efficient for deep exploration.